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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

ME2235-1 - DIGITAL TECHNOLOGIES INCLUDING CPS, IIOT & CLOUD IN MANUFACTURING
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full question from Unit - I and II and One full Question from Unit – III.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following technologies is NOT part of digital manufacturing?
A) CNC
B) AR
C) IIoT
D) Steam Engine
2. Industry 4.0 was first conceptualized in which country?
A) USA
B) Japan
C) Germany
D) India
3. Which CAD modeling type provides a skeletal view of objects?
A) Solid modeling
B) Surface modeling
C) Wireframe modeling
D) Boundary modeling
4. Who pioneered CNC manufacturing in 1949?
A) Henry Ford
B) John T. Parsons
C) Elon Musk
D) Thomas Edison
5. Which revolution introduced programmable logic controllers (PLCs)?
A) First
B) Second
C) Third
D) Fourth
6. Which kernel is used in geometric modeling?
A) RAMI
B) ACIS
C) ERP
D) IIoT
7. What is the main goal of Smart Manufacturing?
A) Reduce labor
B) Increase paperwork
C) Automate operations and improve performance
D) Eliminate machines
8. What is the purpose of ERP in manufacturing?
A) Only design
B) Manage core business processes
C) Replace CAD
D) Control robots
9. Which model is considered more user-friendly than RAMI 4.0?
A) ISA-95
B) SWETI
C) LASFA-LASIM
D) PLM
10. Which technology overlays digital content on the real world?
A) VR
B) AR
C) AI
D) CAM

11. Which of the following is a challenge in CPS design?
 A) Unlimited memory availability B) Real-time performance constraints
 C) Absence of sensors D) Lack of power supply
12. SPI communication is preferred over UART because:
 A) It uses fewer wires B) It is slower than UART
 C) It allows high-speed multi-device communication D) It is asynchronous
13. Which sensor converts sound waves into electrical signals?
 A) Ultrasonic sensor B) Microphone / Sound sensor
 C) Humidity sensor D) Hall-effect sensor
14. Which pin on ARM Cortex is used to read analog sensors like voltage or current?
 A) GPIO digital pin B) ADC pin
 C) PWM output pin D) SPI pin
15. In Arduino Nano, PWM is used to control:
 A) ADC conversion B) Servo motor position
 C) UART communication D) Flash memory
16. Energy monitoring in casting process is primarily used to:
 A) Reduce energy cost B) Predict tool failures
 C) Track raw materials D) Monitor employee attendance
17. Interfacing sensors in CNC machines improves:
 A) Employee productivity B) Tool life and maintenance scheduling
 C) Supplier coordination D) Packaging design
18. AI in Industry 4.0 can help in:
 A) Machine fault prediction B) Energy optimization
 C) Production scheduling D) All of the above
19. FaaS stands for:
 A) Function as a Service B) Factory as a Service
 C) Framework as a Service D) Firmware as a Service
20. Preparing production data for AI optimization involves:
 A) Removing errors and inconsistencies B) Labeling data for ML models
 C) Normalizing and structuring data D) Ignoring missing values

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	What are the elements of digitalization in manufacturing?	8	L*2	1	1
b)	Provide your ideas to convert a conventional machine into a digitalized machine.	8	L3	2	1
2. a)	Analyze the challenges in digital transformation for Industrial Enterprise.	8	L3	1	1
b)	What is a customized industry, and explain with an example?	8	L2	2	1
3. a)	What are the key components of Industry 4.0?	8	L2	1	2
b)	What do you mean by the Integration of the value chain?	8	L2	2	2
Unit – II					
4. a)	List and explain four major challenges in designing Cyber-Physical Systems.	8	L2	3	1

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	b) Explain IIoT applications in packaging industry.	8	L2	4	1
5	a) Explain the working principle of sound sensors and mention two CPS applications.	8	L2	3	1
	b) Explain the working of sensors in CNC machines for predictive maintenance.	8	L2	4	2
6	a) Explain the role of ALU, Control Unit, and Registers in an embedded processor.	8	L2	3	1
	b) Explain IIoT applications in aerospace parts manufacturing.	8	L2	4	2
Unit – III					
7	a) Explain the different building blocks of cloud computing.	8	L2	5	1
	b) Describe PaaS, FaaS, and CaaS models with one industrial example.	8	L2	5	2
8	a) Explain the applications of AI layer in digital factory framework and scope of ML.	8	L2	5	3
	b) Describe the data preparation steps for AI and one real-time industrial example.	8	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
